The instantaneous forward

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Abstract

Basic modeling for forwards in exponential and linear models.

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1 The basics

We consider a forward in presence of a unique risk factor: the equity price, i.e. in presence of deterministic rates and dividends. For the fundamental theorem of asset pricing, the forward $\{F(t,T)\}_{t\geq t_0}$ is a martingale process.

1.1 Exponential model

Exponential models are the natural extension of Black (1976) model. We consider deterministic interest rates and dividend yields (known in t_0) $\{d_t\}_{t\geq t_0}$.
The underlying stock price is the only risk factor present in a forward and their relation at time $t \in [t_0, T]$ is

$$F(t,T) = e^{\big(R(t_0;t,T)-D(t_0;t,T)\big)(T-t)}\; S_t \tag{1}$$

with S_t the stock price at time t and

$$\begin{cases} R(t_0;t,T) & := & \dfrac{1}{T-t}\displaystyle\int_t^T r_s ds \\ D(t_0;t,T) & := & \dfrac{1}{T-t}\displaystyle\int_t^T d_s ds \end{cases} .$$

where $\{r_t\}_{t\geq t_0}$ are the deterministic instantaneous interest rates.
In exponential models, we model the forward at time t with expiry T as

$$F(t,T) =: F(t_0,T)\; e^{f_t} \quad \forall t \in [t_0,T] \;\; , \tag{2}$$

and we call f_t the (exponential) instantaneous forward. Let us observe that $f_{t_0} = 0$.

Lemma 1. *The instantaneous forward f_t defined in* (2) *does not depend on the expiry, but only on the time t.*

Proof. Let us consider a forward $F(t,\tilde{T})$ with a maturity $\tilde{T} \in [t,T)$. We have that

$$F(t,\tilde{T}) = e^{\big(R(t_0;t,\tilde{T})-D(t_0;t,\tilde{T})\big)(\tilde{T}-t)}\; S_t = F(t_0,\tilde{T})\; e^{\tilde{f}_t} \;\; .$$

where $\tilde{f}_t$ in principle could be different from f_t. Using (1) and (2), we can compute the ratio

$$\frac{F(t,T)}{F(t,\tilde{T})} = e^{\big(R(t_0;\tilde{T},T)-D(t_0;\tilde{T},T)\big)(T-\tilde{T})} = \frac{F(t_0,T)}{F(t_0,\tilde{T})} e^{f_t-\tilde{f}_t} \quad . \tag{3}$$

Because the second term in (3) is a deterministic function of time (and equal to $F(t_0,T)/F(t_0,\tilde{T})$), then f_t and $\tilde{f}_t$ must be equal □

Let us observe that e^{f_t} is a martingale process. This is a consequence of the forward being a martingale. Thus, due to definition (2), $\mathbb{E}_0[e^{f_t}] = 1 \;\; \forall t \in [t_0,T]$.

1.2 Linear model

Linear models are the natural extension of Bachelier (1900) model. We consider deterministic interest rates and absolute dividends.

In presence of deterministic absolute dividends, the forward at time t with expiry T is

$$F(t,T) = e^{R(t_0;t,T)\,(T-t)}\ S_t\ -\int_t^T Q_s^T ds \tag{4}$$

with S_t the stock price at time t and Q_s^T absolute dividend paid at time s and capitalized up to T.

In linear models, we model the forward as

$$F(t,T) =: F(t_0,T)\ +\ f_{t,T} \quad \forall t \in [t_0,T]\ , \tag{5}$$

We call $f_{t,T}$ the linear instantaneous forward. Because the forward $\{F(t,T)\}_{t\geq t_0}$ is a martingale process, even the linear instantaneous forward is martingale.

Lemma 2. *The linear instantaneous forward $f_{t,T}$ defined in* (5) *for different expiries differ only for a multiplicative constant,*

$$f_{t,T} = e^{\int_{\tilde{T}}^T r_s ds} f_{t,\tilde{T}} \qquad \forall\, \tilde{T} \in [t,T)$$

Proof. The proof is similar to the exponential case. Let us consider a forward $F(t,\tilde{T})$ with a maturity $\tilde{T} \in [t,T)$. We have that

$$F(t,\tilde{T}) = e^{R(t_0;t,\tilde{T})\,(\tilde{T}-t)}\ S_t\ -\int_t^{\tilde{T}} Q_s^{\tilde{T}} ds = F(t_0,\tilde{T})\ +\ f_{t,\tilde{T}}\ \ .$$

Using (4) and (5), we can compute the difference

$$\begin{aligned} F(t,T) - e^{\int_{\tilde{T}}^T r_s ds}\, F(t,\tilde{T}) &= -\int_t^T Q_s^T ds + \int_t^{\tilde{T}} Q_s^T ds = -\int_{\tilde{T}}^T Q_s^T ds = \\ \left(F(t_0,T) - e^{\int_{\tilde{T}}^T r_s ds}\, F(t_0,\tilde{T})\right) &+ \left(f_{t,\tilde{T}} - e^{\int_{\tilde{T}}^T r_s ds} f_{t,\tilde{T}}\right) = -\int_{\tilde{T}}^T Q_s^T ds + \left(f_{t,\tilde{T}} - e^{\int_{\tilde{T}}^T r_s ds} f_{t,\tilde{T}}\right)\ , \end{aligned} \tag{6}$$

that proves the lemma. It is a consequence deterministic absolute dividends ☐

References

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Black, F., 1976. The pricing of commodity contracts, *Journal of financial economics*, 3 (1-2), 167–179.

Notation

Symbol	Description
$\mathbb{E}_s[\bullet]$	$\mathbb{E}_s[\bullet \vert \mathscr{F}_s]$ where $\mathscr{F}_s$ is the natural filtration
$F(t,T)$	Forward price with expiry in T and valued at time $t \in [t_0, T]$
f_t	(exponential) instantaneous forward
$f_{t,T}$	linear instantaneous forward
$R(t_0; t, T)$	forward zero rate
r_t	instantaneous rate
d_t	dividend yield
Q_s^T	absolute dividend paid at time s and capitalized up to T